

# Interference Filter External Cavity Diode Laser (ECDL) with a Cateye Configuration

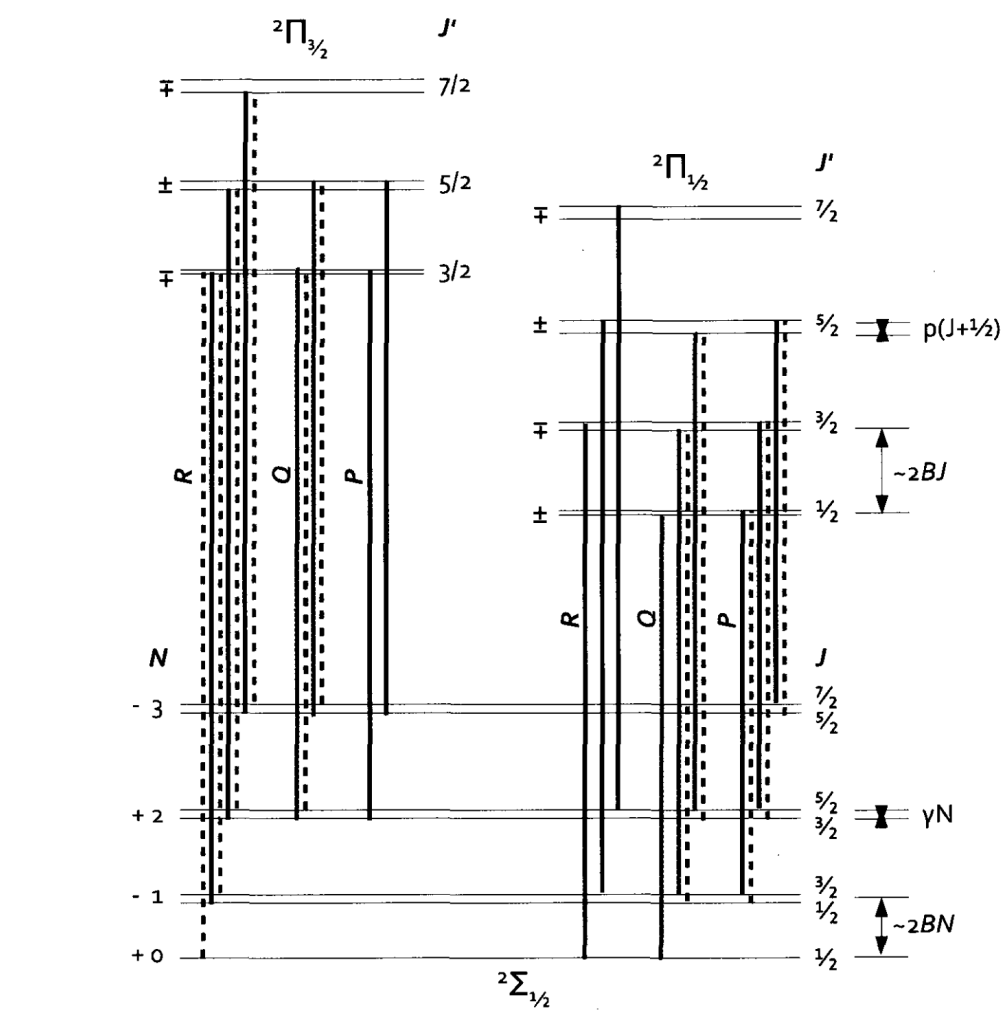
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## Goals and Motivation

### ZOMBIES: Use <sup>137</sup>Ba<sup>19</sup>F to measure nuclear spin-dependent parity violation (NSD-PV) effect

#### Rotational Cooling (Pump population into single hyperfine level)

- Laser optically pumps <sup>137</sup>Ba<sup>19</sup>F molecules from  $X^2\Sigma(v=0; N^P=0^+, 1^-, 2^+)$  to  $A^2\Pi_{1/2}(J^P=5/2^+)$ , leaving  $X^2\Sigma(v=0; N^P=1^-, J=3/2)$  hyperfine level unaddressed.
- Spontaneously decay back to ground state, accumulate population to  $X^2\Sigma(v=0; N^P=1^-, J=3/2)$ .
- Simulation shows **×30** gain in the  $X^2\Sigma(v=0; N^P=1^-, J=3/2)$  population.



#### Laser Requirements

- Narrow linewidth (~100 kHz)  
*Similar Level of precision would mean measuring earth-moon distance with an uncertainty of 1mm!*
- Wide tunability (~30GHz)
- High stability.
- Power output (~100mW)
- Ease of use, small size, reproducible.

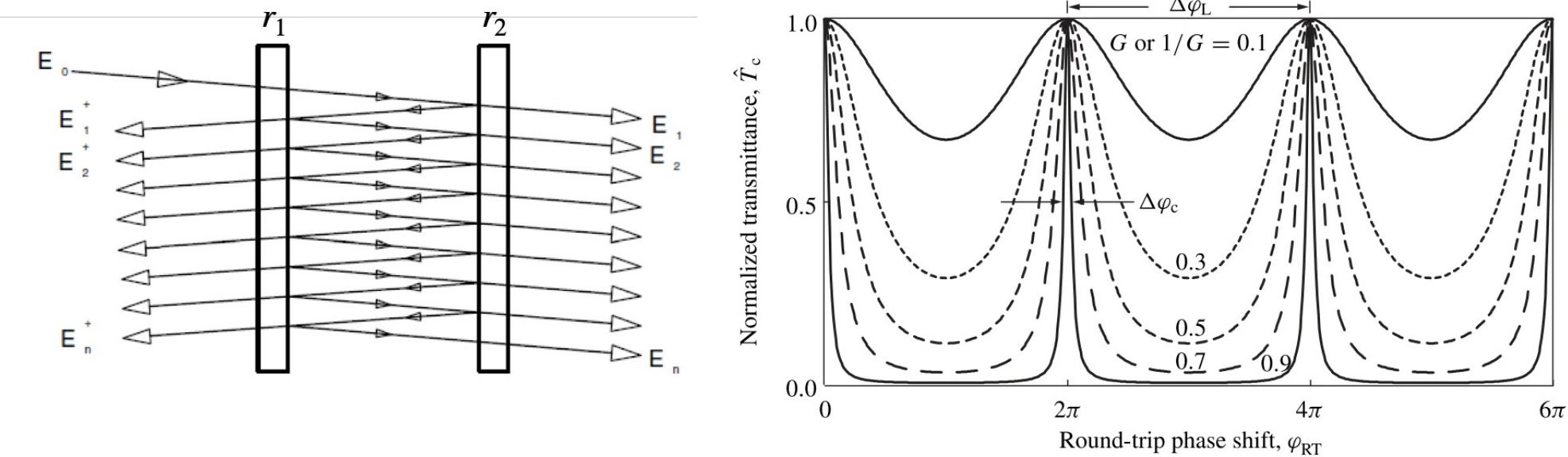
## Theoretical Basis

### Fabry-Perot Cavity

A Fabry–Perot cavity is an optical resonator consisting of **two parallel, highly reflective mirrors** that trap light through multiple reflections, creating standing waves at specific resonant frequencies.

$$T_{cavity}(\nu, L) = \frac{1}{1 + F \sin^2 \delta(\nu)}$$

where  $F$  is the coefficient of **finesse** for the cavity.  $\delta(\nu) = 2\pi nLv/c$ ,  $L$  is the **cavity length**, and  $n$  is the refractive index of the cavity medium.

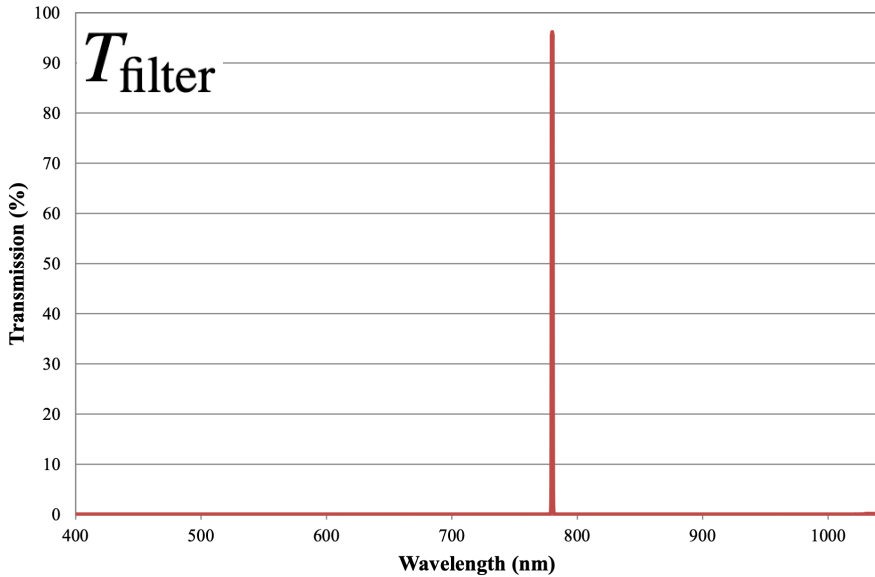


### Ultra-narrow Bandpass Interference Filter

An optical filter that selectively transmits light of certain wavelengths (~1nm) by exploiting constructive and destructive interference between multiple thin-film layers.

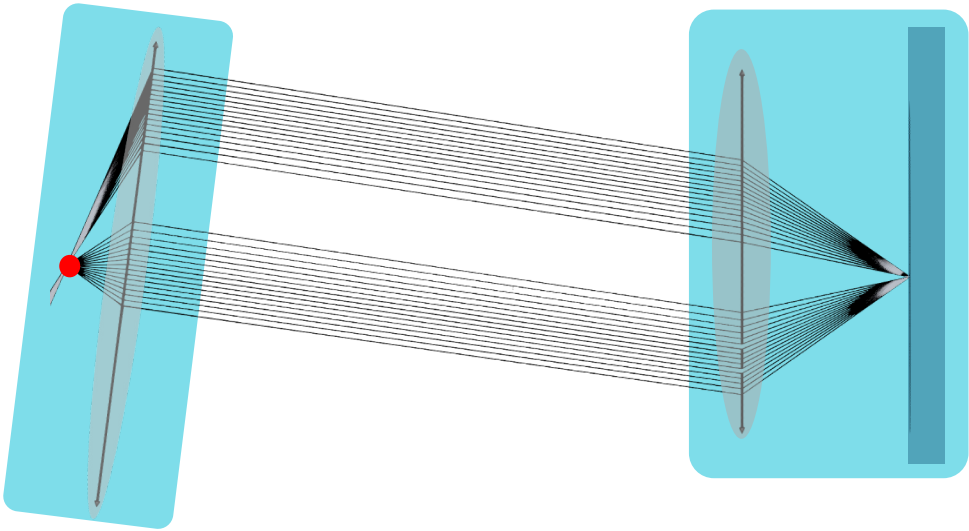
The transmission window wavelength shift with incident angle  $\theta$ :

$$\lambda(\theta) = \lambda_0 \sqrt{1 - \left( \frac{\sin(\theta)}{n} \right)^2}$$



### Cateye Retroreflector

Consists of a **lens-mirror combination (Cateye)** and a **lens-source pair** aligned to face each other, with the mirror/light source positioned at the focal plane/point of the lens. This set up ensures any collimated beam from the lens-source pair is reflected back to the source, regardless of their relative angle (provided the beam strikes the Cateye).



#### Advantages:

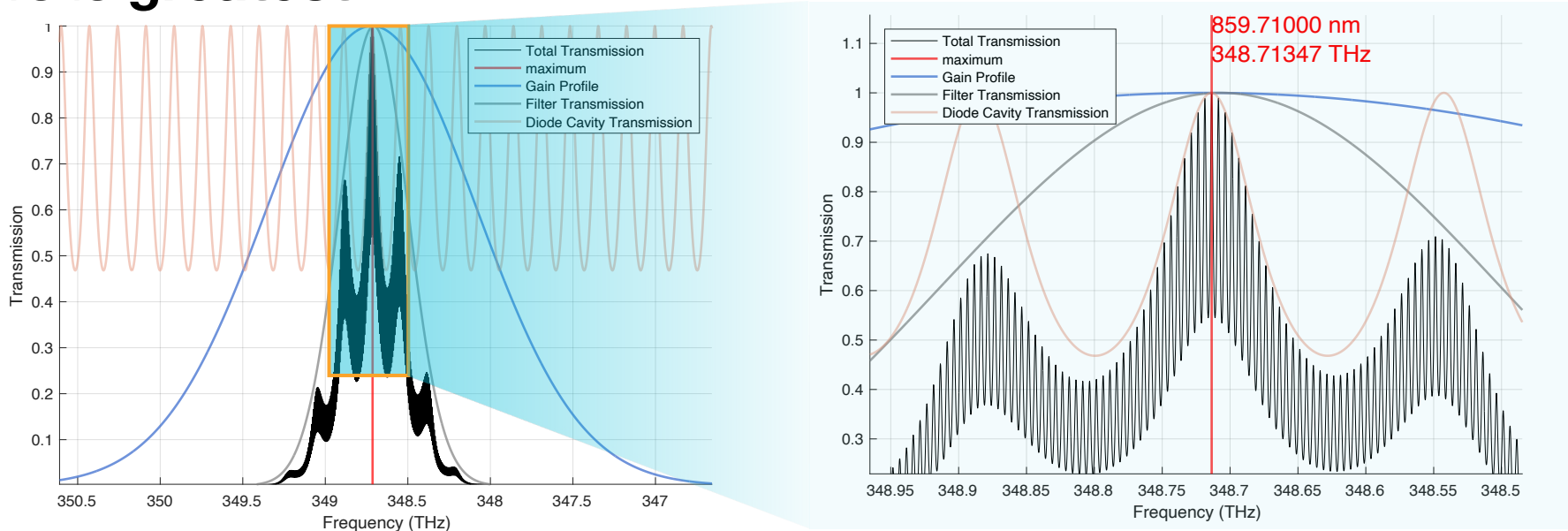
- Easy to align
- Mechanically stable

### Laser Output Frequency

The output frequency  $\nu$  of an ECDL depends on a combination of frequency-dependent gain and loss factors, and the product of these factors gives the total transmission function:

$$T_{total} = G_D T_D T_{cavity} T_{filter}$$

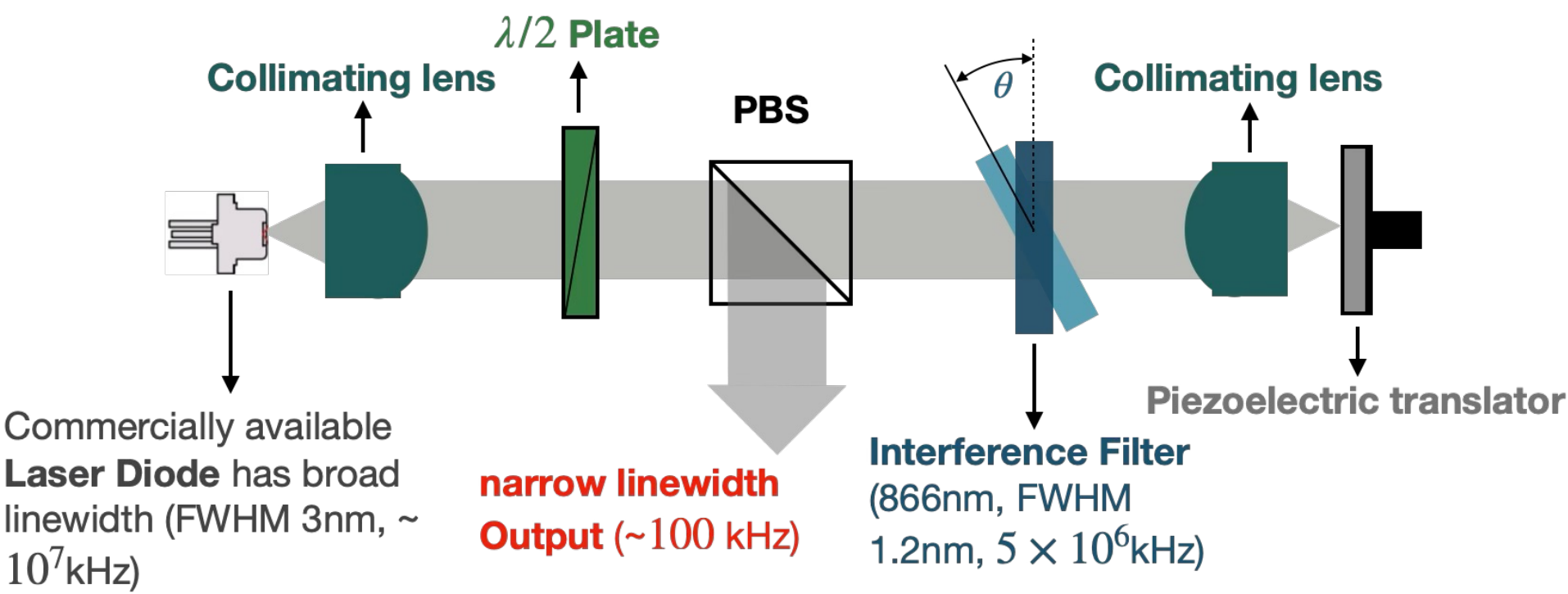
The laser will oscillate at the frequency for which the product of these factors is **greatest**.



## Design

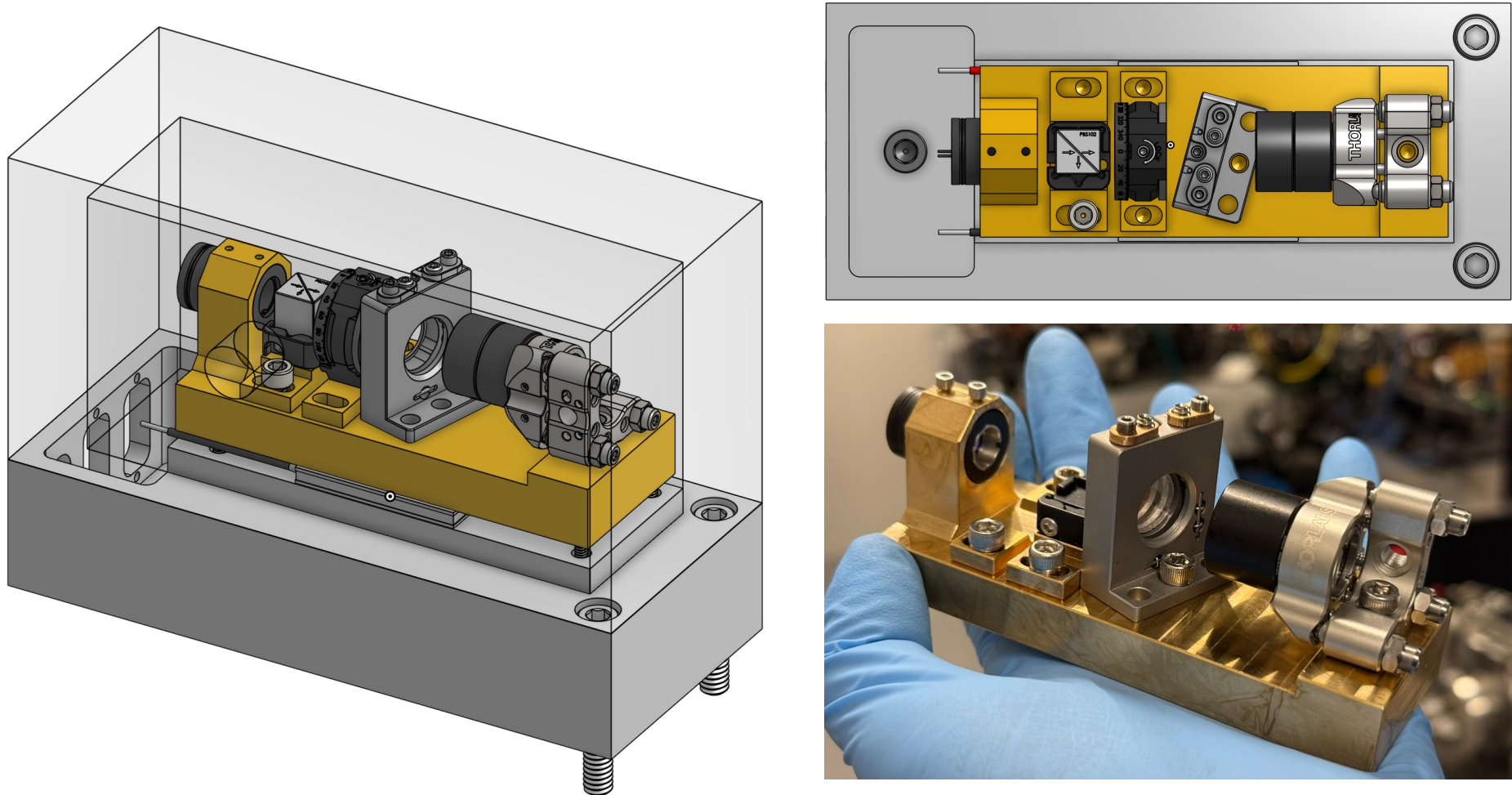
### Design Principles

- $\lambda/2$  waveplate + PBS allows tuning the feedback intensity
- Self-aligning Cateye reflection geometry, mechanically robust
- Frequency coarse tuning by changing diode temperature (100GHz)
- Moderate tuning by changing laser diode current (10GHz)
- Fine tuning by changing piezo voltage (<1GHz)



### CAD

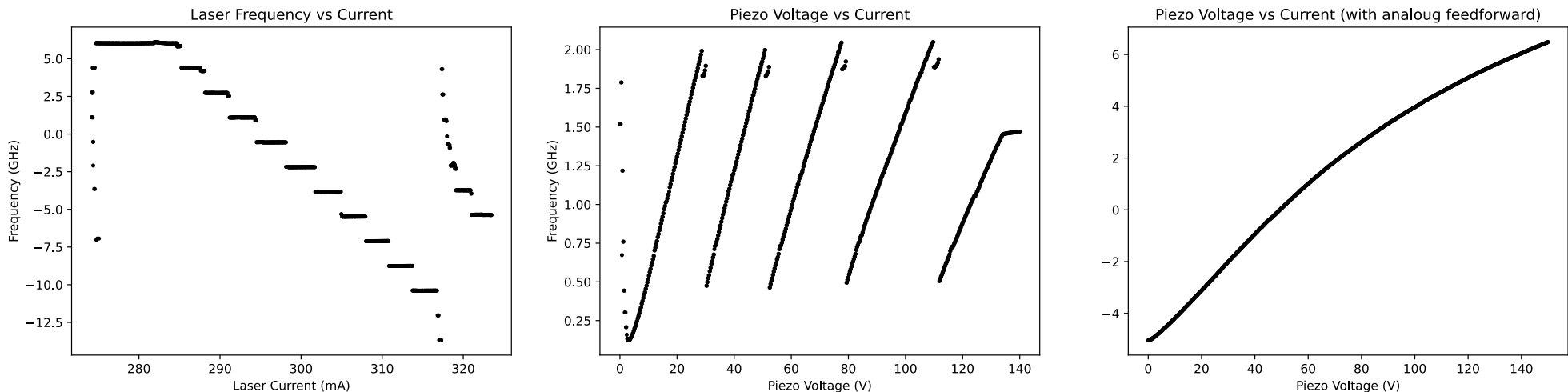
- Precise pitch and roll control of Cateye configuration allows finer tuning of alignment
- 1-inch-thick acrylic lid ensures system's thermal stabilization
- Slotted PBS and  $\lambda/2$  waveplate brass spacer allows adjustment of their position
- Brass base provides good thermal conductivity



## Result

### Diode Current Scan, Piezo Voltage Scan, and Feedforward

- Without feedforward (scanning current and voltage simultaneously at correct ratio), scanning current gives discrete steps, scanning voltage gives ~1.5GHz MHFTR, consistent with simulation
- With feedforward, MHFTR ~12GHz



## Outlook

Transfer Cavity for frequency stabilization and frequency shifting laser

